

FULL STACK DEVELOPMENT WITH MERN

PROJECT DOCUMENTATION

Online Complaint Registration System

Prepared By: M.NEHA REDDY

Project Documentation for Academic Submission

1. Introduction

The Online Complaint Registration System is a web-based platform developed using the MERN Stack. It provides a centralized mechanism for users to register complaints, track their progress, and receive updates. The system reduces paperwork, increases transparency, and improves complaint resolution efficiency.

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2. Project Overview

The project aims to digitize complaint management. Users can submit complaints online while administrators manage and resolve them through a dashboard. The platform supports authentication, complaint tracking, notifications, and reporting.

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3. Problem Statement

Traditional complaint handling methods rely on paper forms, emails, and manual processes. These approaches lead to delays, poor tracking, and lack of accountability. A centralized digital system is required.

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4. Objectives

- Digitize complaint management
- Improve transparency
- Enable complaint tracking
- Reduce manual effort
- Improve communication
- Increase user satisfaction

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5. System Architecture

The system follows a three-tier architecture:

Presentation Layer (React.js)

Business Logic Layer (Node.js + Express.js)

Database Layer (MongoDB Atlas)

Flow: User → Frontend → Backend APIs → Database → Response

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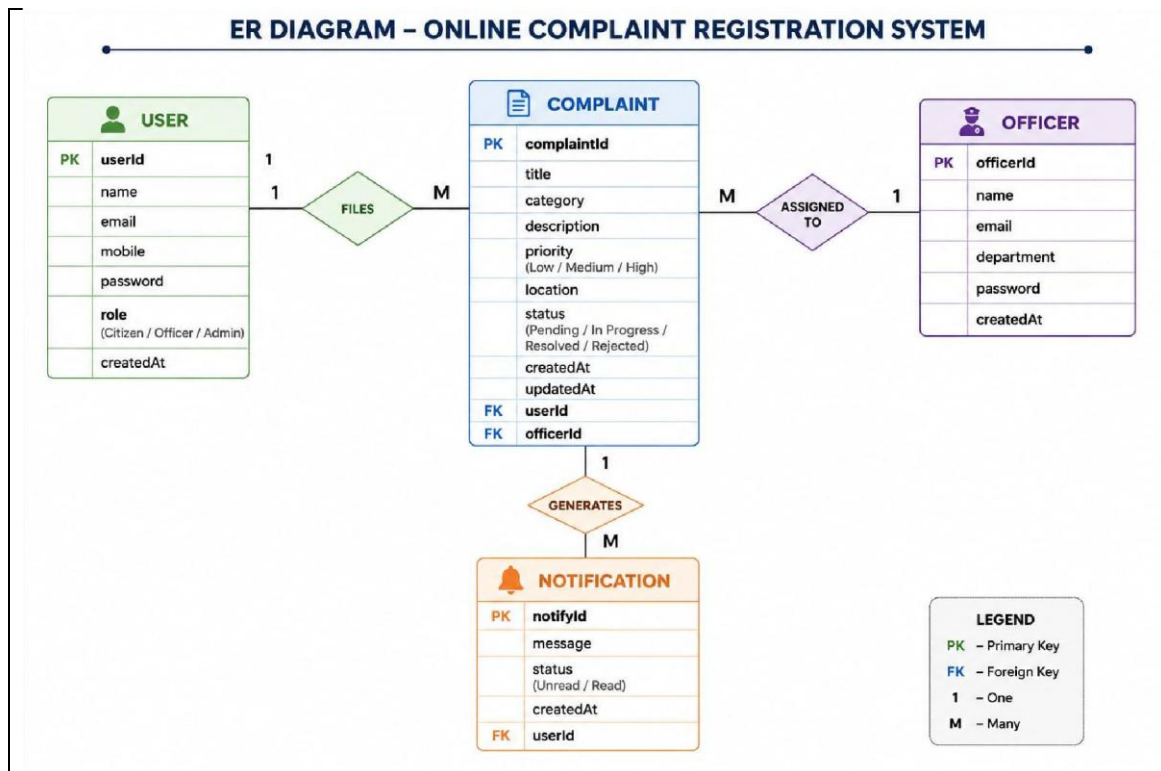
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6. Technology Stack

Frontend: React.js, Vite, Axios, React Router DOM

Backend: Node.js, Express.js

Database: MongoDB Atlas, Mongoose

Authentication: JWT, bcryptjs

Deployment: Vercel, Render

Version Control: Git, GitHub

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7. Setup Instructions

Install Node.js, Git, and MongoDB Atlas account.

Clone repository.

Install dependencies using npm install.

Configure .env variables.

Run frontend and backend servers.

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8. Folder Structure

Client: components, pages, services, routes

Server: config, controllers, middleware, models, routes This structure ensures maintainability and scalability.

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9. Application Workflow

User Registration → Login → Submit Complaint → Complaint Review → Status Update → Resolution → Notification

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10. API Documentation

Authentication APIs:

POST /register

POST /login

Complaint APIs:

POST /complaints

GET /complaints

PUT /complaints/:id

Admin APIs:

GET /all-complaints

PUT /status-update

Detailed Discussion: Authentication APIs:

POST /register

POST /login

Complaint APIs:

POST /complaints

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Complaint APIs:

POST /complaints

GET /complaints

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GET /all-complaints

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11. Authentication & Security

JWT Authentication, bcrypt password hashing, protected routes, role-based access control, secure API communication.

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12. User Interface

The application includes Login Page, Registration Page, User Dashboard, Complaint Submission Form, Complaint Tracking Page, Admin Dashboard, and Notification Center.

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13. Testing

Functional Testing, API Testing, Integration Testing, and User Acceptance Testing were performed. All major modules passed successfully.

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14. Deployment

Frontend: <https://online-complaint-registration.vercel.app/>

Backend: <https://online-complaint-registration.onrender.com>

Database: MongoDB Atlas

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15. Known Issues

Email notifications and SMS integration are not yet implemented. Mobile optimization can be improved further.

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16. Future Enhancements

AI-based complaint categorization, analytics dashboard, SMS alerts, mobile application, multilingual support, and automated complaint routing.

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17. Conclusion

The Online Complaint Registration System successfully demonstrates the implementation of a full-stack MERN application. It improves transparency, efficiency, and user satisfaction while providing a scalable architecture for future growth.

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Live Project Links

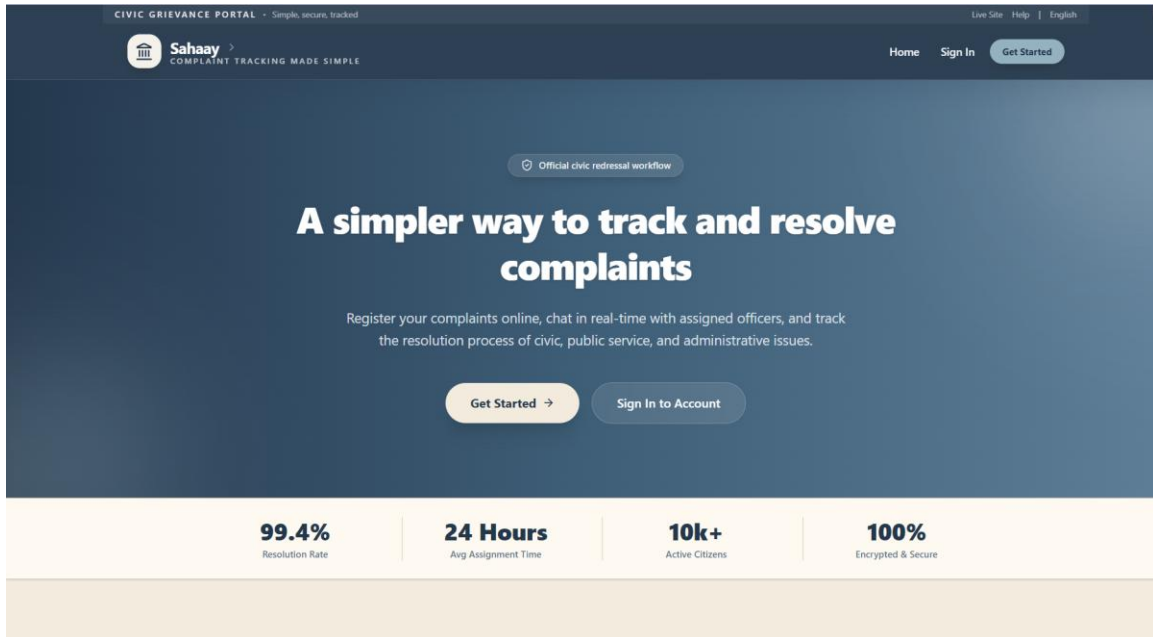
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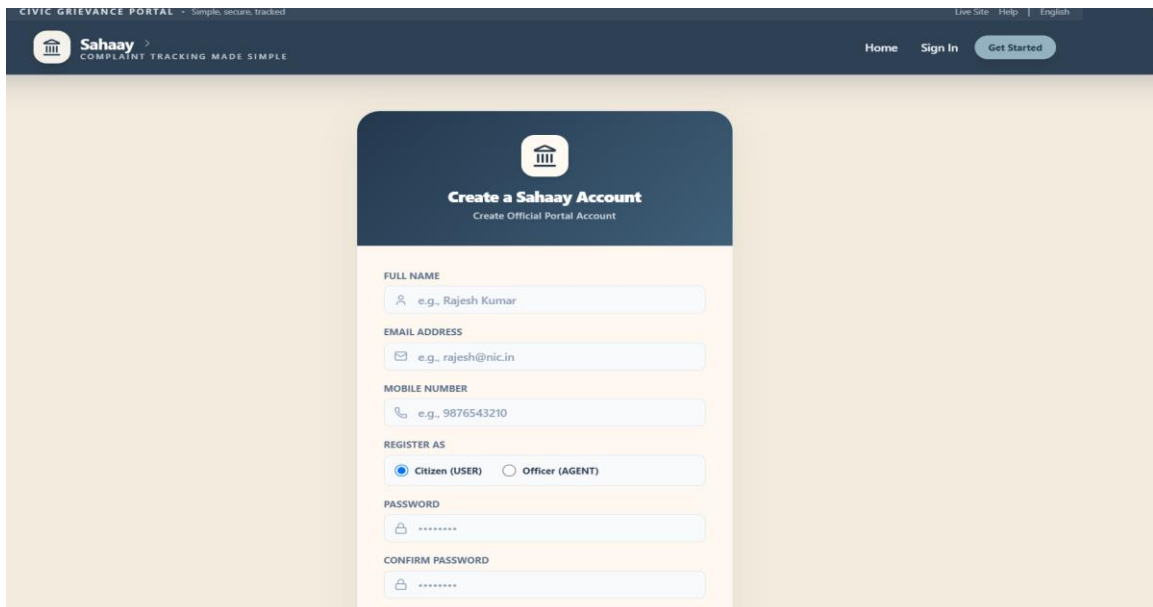
GitHub: <https://github.com/nehareddy0705/online-complaint-registration>

ONLINE COMPLAINT REGISTRATION SYSTEM

Home Page



Registration Page



The screenshot shows the registration page of the Sahaay Civic Grievance Portal. It features a central form titled 'Create a Sahaay Account' with fields for personal information and account creation options.

CIVIC GRIEVANCE PORTAL • Simple, secure, tracked

Live Site Help English

Sahaay COMPLAINT TRACKING MADE SIMPLE

Home Sign In Get Started

Create a Sahaay Account

Create Official Portal Account

FULL NAME
e.g., Rajesh Kumar

EMAIL ADDRESS
e.g., rajesh@nic.in

MOBILE NUMBER
e.g., 9876543210

REGISTER AS
☒ Citizen (USER) ☐ Officer (AGENT)

PASSWORD

CONFIRM PASSWORD

User Dashboard

CIVIC GRIEVANCE PORTAL · Simple, secure, tracked

Sahaay

COMPLAINT TRACKING MADE SIMPLE

Live Site · Help · English

Neha Reddy

USER

Logout

N

Neha Reddy

mittabaneha705@gmail.com

USER PORTAL

CITIZEN SERVICES

My Grievances

File New Grievance

My Grievance Dashboard

Submit complaints, interact with assignment officers, and monitor updates.

File New Grievance

TOTAL FILED

1

PENDING

1

ACTIVE TASKS

0

RESOLVED

0

Search by title, category...

ALL · PENDING · ACTIVE / ASSIGNED · RESOLVED · REJECTED

WATER SUPPLY & SANITATION

Pending

MEDIUM Priority

water leakage

water leak just like neet paper leak

Filed: 20 Jun 2026

View Details →

File New Grievance

CIVIC GRIEVANCE PORTAL · Simple, secure, tracked

Sahaay

COMPLAINT TRACKING MADE SIMPLE

Live Site · Help · English

Neha Reddy

USER

Logout

N

Neha Reddy

mittabaneha705@gmail.com

USER PORTAL

CITIZEN SERVICES

My Grievances

File New Grievance

File Official Grievance

Submit details below to register your grievance. A tracking number will be generated immediately.

COMPLAINT TITLE *

Briefly state the issue (e.g., Water leakage in sector 4 main road)

DEPARTMENT / CATEGORY *

-- Choose Department --

ESTIMATED URGENCY / PRIORITY *

MEDIUM

FULL NARRATIVE DESCRIPTION *

Please explain the issue in detail. Mention when it started, what has been affected, and any specific support details.

INCIDENT ADDRESS / LOCATION DETAILS *

House / Street / Landmark details

CITY / TOWN *

e.g., Indore

STATE / PROVINCE *

-- Choose State --

POSTAL CODE / PINCODE (6-DIGITS) *

e.g., 452001